

Applied A.I. Solutions Development Program
Applied Mathematical Concepts for Deep Learning

Term Project

Predict Calorie Expenditure

GROUP 7

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Agenda

- Introduction: Friba Hussainyar
- Dataset Overview: Gavriel Kirichenko
- Feature Engineering & Selection: Callum Arul
- Model Implementation & Hyperparameter Tuning: Hsi-Teng Liu
- Model Evaluation: ABdul-Rasaq Omisesan
- Demo: Diparshan Bhattarai
- Next Step: Chia-Wei Chang
- Conclusion: Bikash Giri
- Lessons to Learn

Introduction

- Kaggle Competition
- 3903 participant
- we are 43
- Link: [Predict Calorie Expenditure | Kaggle](#)
- Our objective was to develop a DL model that accurately predicts calorie expenditure based on physiological and activity data.
- The challenge was to predict how many calories a person burns during physical activity using features such as age, weight, gender, height and heart rate. Accurate calorie estimation can help people better manage their fitness, nutrition, and health goals.
- For the prediction, we used a Multi-Layer Perceptron (MLP) built with TensorFlow/Keras and GPU. We fine-tuned it through several experiments and finally selected the best model using cross-validation. Our model achieved excellent results with an R^2 of 0.9954 and a very low RMSLE of 0.0623.

Dataset Overview

The dataset comprises anonymized workout session data, with each row representing a single session. The primary objective is to predict the number of calories burned during these sessions.

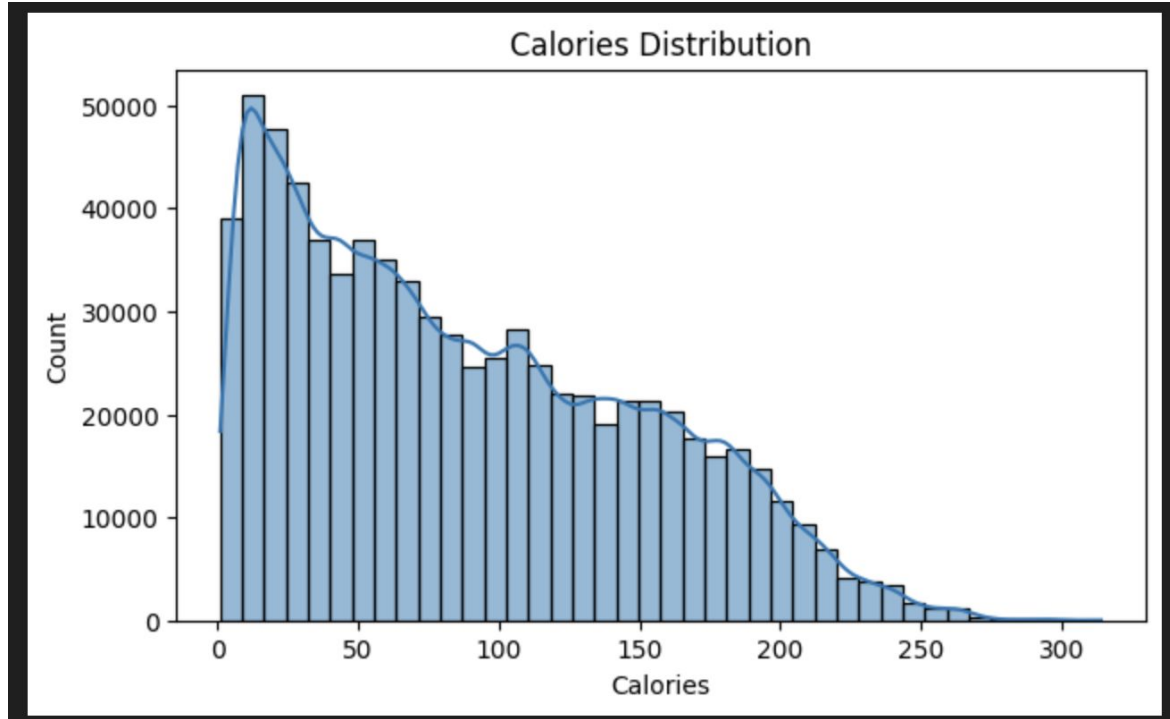
Key Features:

- **Numerical Variables:** Age, Height, Weight, Duration, Heart_Rate, Body_Temp
- **Categorical Variables:** Sex
- **Target Variable:** Calories

Evaluation Metric:

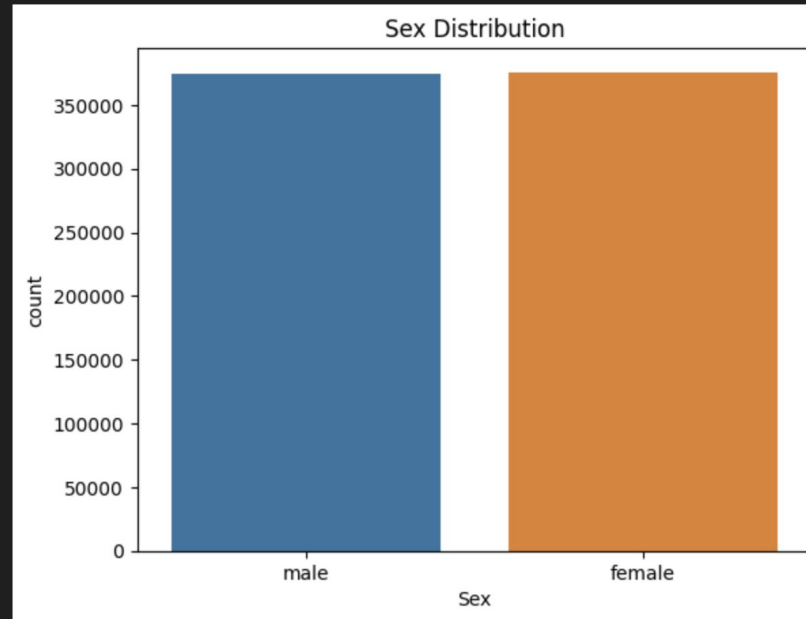
The evaluation metric for this competition is Root Mean Squared Logarithmic Error (RMSLE), which penalizes underpredictions more than overpredictions and is suitable for targets that span a wide range.

EDA - Calories Distribution

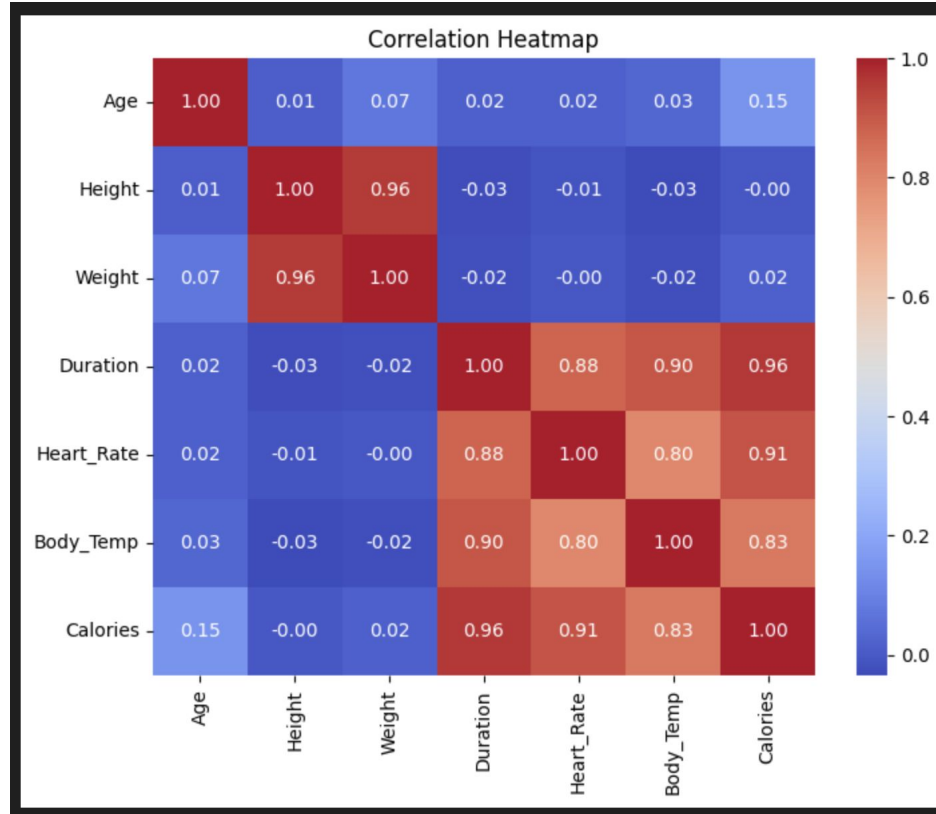


EDA - Sex Distribution

```
Sex value counts:  
Sex  
female    375721  
male      374279  
Name: count, dtype: int64
```



EDA - Correlation Heatmap



EDA - Removing Outliers

- No missing values
- Outliers beyond $1.5 \times \text{IQR}$ were capped at the nearest whisker.

Feature Engineering & Selection

Code: [Dense notebook5a7e8f4039](#)

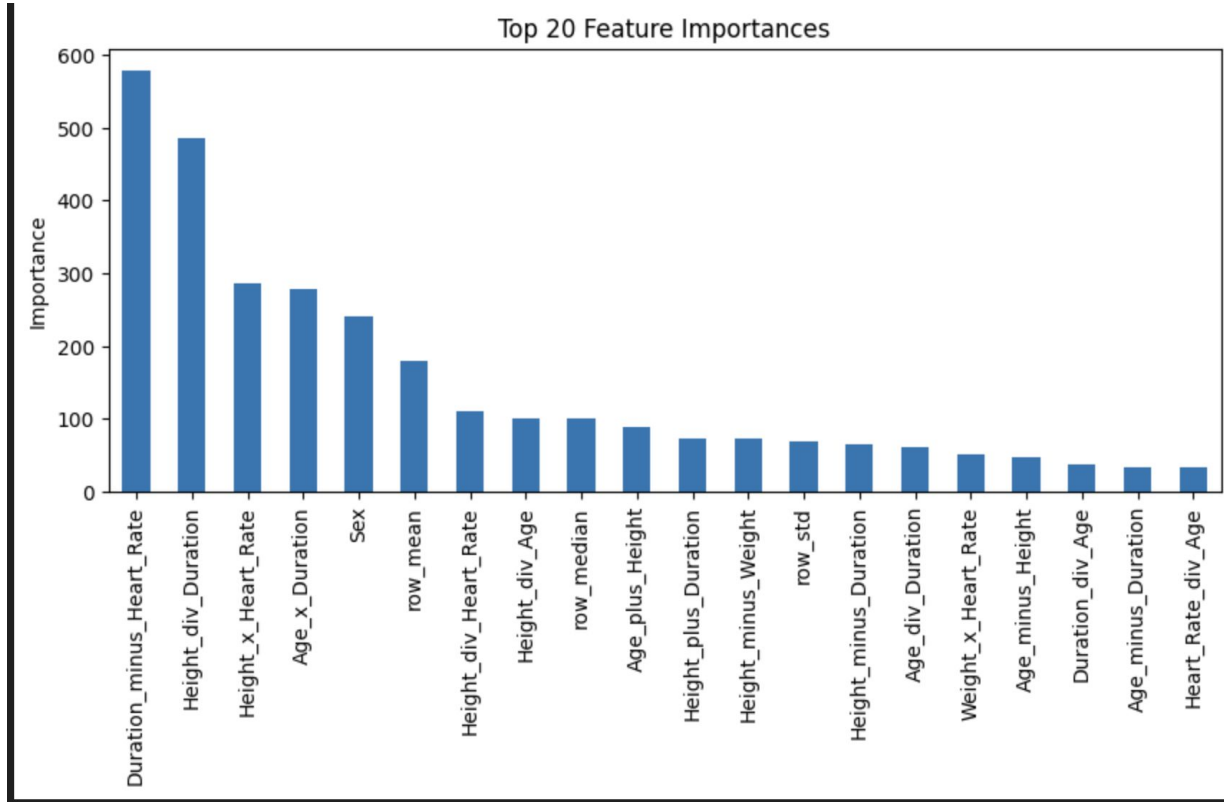
Added new features:

- Cross Terms (e.g. `Weight × HeartRate`)
- Interaction Features - Sum, difference, and ratio terms
- Row-wise Stats (mean, std, max, etc.)

Feature Selection:

- Dropped features with **correlation > 0.95** (reduces redundancy)
- Selected **Top 100** via LightGBM feature importance
- Visualized **Top 20** to gain insights into key predictors

Feature Engineering & Selection



Model Implementation & Hyperparameter Tuning

First Attempt: A Simple Approach with MinMaxScaler

Layer 1: 6 nodes with tanh activation

Layer 2: 4 nodes with ReLU activation

Learning Rate: 0.001

Kaggle Public Score = 0.06064

Model Implementation & Hyperparameter Tuning

Second Try with Help from Optuna

Layer 1: 555 nodes with ReLU activation, Batch Normalization, Dropout of 0.277

Layer 2: 277 nodes with ReLU activation, Batch Normalization, Dropout of 0.277

Learning Rate: 0.0001277

Kaggle Public Score = 0.05896

```
hidden = trial.suggest_int("hidden", 128, 1024)
dropout = trial.suggest_float("dropout", 0.1, 0.5)
lr = trial.suggest_float("lr", 1e-4, 5e-3)
```

MODEL EVALUATION

Cross Validation

5-Fold KFold CV on the full training set, with random shuffling and a fixed seed for reproducibility



Loss Curve Diagnostics

Fast convergence
curves overlap closely
No sign of divergence



Predicted Vs True Scatter

This plot sits at approximately 45 degrees diagonal
Overall tight cluster confirming accuracy & precision



Core Metrics

Root mean Squared Error- 0.06230
Mean Absolute Error- 2.528 kcal
R Squared - 0.9954



Residual Analysis

Residuals scatter symmetrically around zero
No obvious non linear pattern

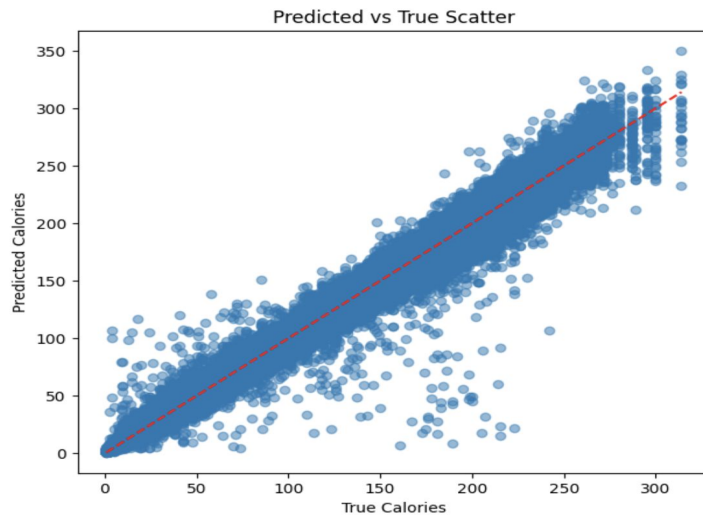
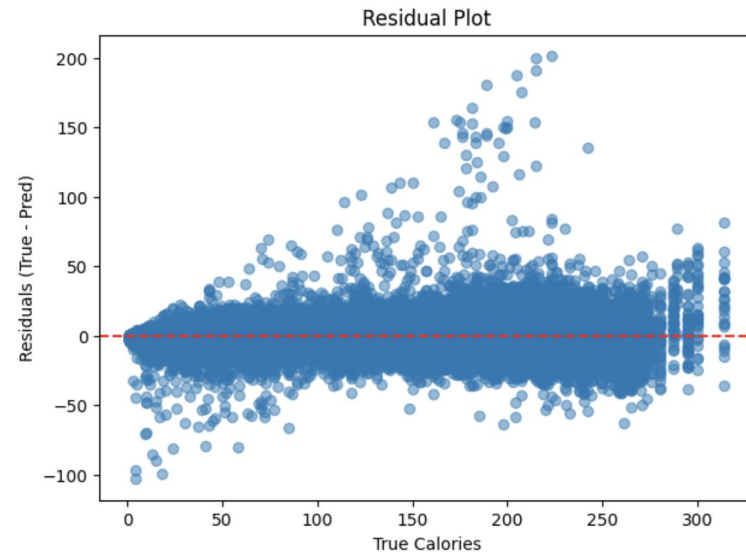
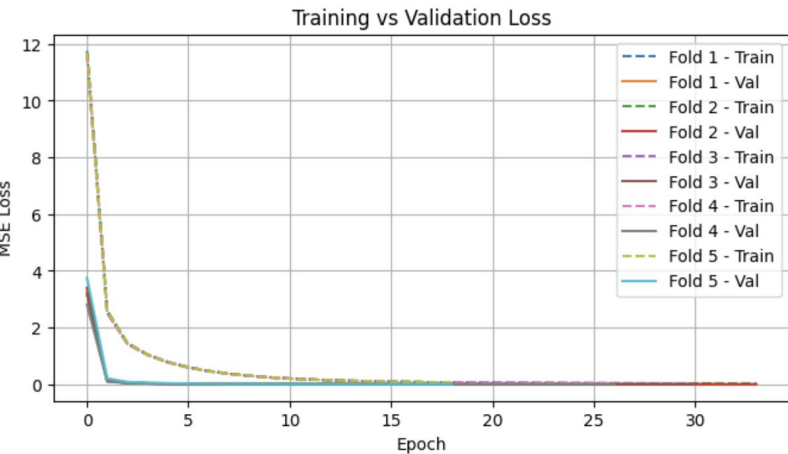


Robust Binning

Three quantile bin split
Accuracy - 97.8%
Macro F1 - 0.978
Precision - 0.978



Model Evaluation Contd



Demo

Code: [link](#)

Input User Features

Sex
male

Age
25

Height (cm)
170

Weight (kg)
70

Duration (min)
30

Heart Rate
120

Body Temp (C)
37

 Model Training Reference

[Kaggle Notebook: Dense Notebook](#)

Predict Calorie Expenditure

Predict

 **Predicted Calorie Expenditure: 228.17 cal**

Model Evaluation Metrics (from cross-validation):

- RMSLE: 0.0623
- MAE: 2.528
- R^2 : 0.9954

Scores are based on out-of-fold (OOF) cross-validation on the training set.

Classification-style Evaluation (Calories Binned):

- Accuracy: 0.978
- F1 Score: 0.978
- Precision: 0.978
- Recall: 0.978

(Binning: [1, 47, 113, 314] on validation set)

Next Step

What we did

1. Feature Engineering

- Added cross-product features (e.g., Age × Height).
- Created interaction features (sum, difference, division).
- Generated row-level statistics (mean, std, max, etc.).

2. Feature Selection

- Removed highly correlated features (correlation > 0.95).
- Used LightGBM to select top 100 important features.

3. Data Preprocessing

- Standardized numerical features using StandardScaler.

4. Model Optimization with Optuna

- Defined objective function for hyperparameter tuning of MLP (hidden size, dropout, learning rate).
- Used 5-fold cross-validation to evaluate performance.
- Selected best hyperparameters after 20 trials.

5. Model Training & Prediction

- Trained final MLP model with best parameters using 5-fold CV.
- Saved model and preprocessing artifacts (scaler, top features).
- Generated test predictions by averaging across folds.

How to Improving Competition Score:

Level 1: Advanced Modeling Strategy: Model Stacking

- To achieve a higher score in the competition, we need to go beyond using a single model.
- Currently, we're relying on one model, but we can improve performance through model stacking combining multiple models such as CatBoost, XGBoost, and LightGBM.
- By applying ensemble techniques like voting or blending, we can make our predictions more accurate, stable, and competitive.

Level 2: Stacking Public Submissions for Higher Scores

- As we observed from other teams in the competition, some teams achieved higher scores by stacking public submission data from other Kaggle notebook.
- By identifying groups with higher leaderboard scores, we can:
 - Collect 10+ strong public submissions
 - Ensemble their predictions using techniques like median voting
 - Improve the robustness and accuracy of our final prediction

Conclusion

- Successfully developed a deep learning model (MLP) to predict calorie expenditure using physiological and activity data.
- Achieved high accuracy with $R^2 = 0.9954$ and RMSLE = 0.0623 using optimized hyperparameters from Optuna.
- Feature engineering (cross terms, interaction features, row-wise stats) and proper preprocessing significantly improved performance.
- Applied 5-fold cross-validation to ensure robustness and avoid overfitting.
- Demonstrated that smarter features and model tuning often outperform more complex models.
- Future improvements include ensemble strategies like model stacking and blending predictions from public submissions.
- Overall, gained practical experience in end-to-end deep learning project implementation for a real-world use case.

Lessons to Learn

- **Bikash Giri (101575097)** - I learned what streamlit is and how to use it as a front end for the machine learning model to make it more interactive for the end user. I figured what h5 file is and what is the process to integrate it into the frontend.
- **Chia-Wei Chang (101570243)** - I learned how to train a dense deep learning model using Optuna to find the optimal hyperparameters, including the number of nodes and How to get high score during Kaggle competition.
- **Diparshan Bhattarai (101494737)** - I learned how to prepare data for modeling by removing redundant features, selecting the most important ones using LightGBM, and standardizing inputs to improve model performance and accuracy.
- **Hsi-Teng Liu (101576074)** - Sometimes it's worth considering a simpler mode which has slightly worse performance, but needs significant less resources when training and inferencing.
- **Gavriel Kirichenko (101119609)** - I learned that deep learning isn't always the best solution in fact, in our case, smaller models like XGBoost performed better.
- **Abdul-Rasaq Omisesan (101571156)** - I learned that thoughtful feature engineering and 5-fold cross-validation can beat simply building bigger networks.
- **Callum Arul (101585383)** - I learned that you can dramatically improve model performance not just by tuning models, but by engineering smart features
- **Friba Hussainyar (101591222)** - I learned how to use the streamlit and connect the h5 model file to the front-end.